

Q&A

Open Addressing, Reductions, and 3-Coloring

1 Question 3

Question

Insert the keys

10, 22, 31, 4, 15, 28, 17, 88, 59

into a hash table of length $m = 11$ using open addressing with auxiliary hash function $h'(k) = k$. Illustrate the result using:

- quadratic probing with $c_1 = 1$ and $c_2 = 3$;
- double hashing with $h_1(k) = k$ and

$$h_2(k) = 1 + (k \bmod (m - 1)).$$

Quadratic Probing

Use

$$h(k, i) = (k + i + 3i^2) \bmod 11.$$

Key	Probe sequence until insertion
10	10
22	0
31	9
4	4
15	4, 8
28	6
17	6, 10, 9, 3
88	0, 4, 3, 8, 8, 3, 4, 0, 2
59	4, 8, 7

Final table:

i	0	1	2	3	4	5	6	7	8	9	10
$T[i]$	22	—	88	17	4	—	28	59	15	31	10

Double Hashing

Use

$$h(k, i) = ((k \bmod 11) + i(1 + (k \bmod 10))) \bmod 11.$$

Key	Probe sequence until insertion
10	10
22	0
31	9
4	4
15	4, 10, 5
28	6
17	6, 3
88	0, 9, 7
59	4, 3, 2

Final table:

i	0	1	2	3	4	5	6	7	8	9	10
$T[i]$	22	–	59	17	4	15	28	88	–	31	10

2 Question 4

Question

Prove that the expected number of probes in an unsuccessful search is at most

$$\frac{1}{1 - \alpha}$$

for an open-address hash table with load factor

$$\alpha = \frac{n}{m} < 1,$$

assuming uniform hashing.

Proof

In an unsuccessful search, probes continue until an empty slot is found. Under uniform hashing, the probability that a probed slot is occupied is at most

$$\alpha.$$

Therefore the probability that the first i probes are all occupied is at most

$$\alpha^i.$$

If X is the number of probes, then

$$\Pr\{X > i\} \leq \alpha^i.$$

Using the tail-sum formula for expectation,

$$E[X] = \sum_{i \geq 0} \Pr\{X > i\} \leq \sum_{i \geq 0} \alpha^i = \frac{1}{1 - \alpha}.$$

Write This in Exam

An unsuccessful search needs another probe only if all previous probes hit occupied slots. Under uniform hashing,

$$\Pr\{X > i\} \leq \alpha^i.$$

Thus

$$E[X] = \sum_{i \geq 0} \Pr\{X > i\} \leq \sum_{i \geq 0} \alpha^i = \frac{1}{1 - \alpha}.$$

3 Question 5**Question**

- Define reducibility of a problem X to a problem Y in polynomial time.
- Assume that problem Y can be reduced to problem X in polynomial time. How do we establish that X cannot be solved in polynomial time if Y cannot be solved in polynomial time?

Part a

Problem X is polynomial-time reducible to problem Y , written

$$X \leq_p Y,$$

if there is a polynomial-time computable function f such that for every instance x ,

$$x \in X \iff f(x) \in Y.$$

Thus, an algorithm for Y can be used to solve X after polynomial-time preprocessing.

Part b

Assume

$$Y \leq_p X.$$

Suppose, for contradiction, that X has a polynomial-time algorithm. Then any instance y of Y can be solved as follows:

- compute $f(y)$ in polynomial time;
- run the polynomial-time algorithm for X on $f(y)$;
- answer yes for Y exactly when $f(y)$ is yes for X .

This gives a polynomial-time algorithm for Y , contradicting the assumption that Y cannot be solved in polynomial time. Therefore X cannot be solved in polynomial time.

4 Question 6**Question**

A vertex coloring assigns colors to vertices so that adjacent vertices have distinct colors. A k -coloring uses at most k colors. Prove that the 3-coloring problem is NP-complete.

Exam Proof

First, 3-COLOR is in **NP**. A certificate is an assignment of one of three colors to every vertex. The verifier checks every edge and confirms that its endpoints have different colors. This is polynomial time.

For NP-hardness, reduce from 3-SAT. Given a 3-CNF formula, build a graph using the standard 3-coloring reduction.

- Create three special vertices T, F, B forming a triangle. Their colors represent true, false, and base.
- For each variable x_i , create vertices x_i and $\neg x_i$. Connect them to each other and to B . Therefore they must receive the two colors T and F in opposite order.
- For each clause $(\ell_1 \vee \ell_2 \vee \ell_3)$, attach the standard constant-size clause gadget to the three literal vertices. The gadget is 3-colorable iff at least one of the literals has color T .

If the formula is satisfiable, color each literal vertex according to its truth value. Since every clause has a true literal, every clause gadget can be 3-colored.

Conversely, if the graph is 3-colorable, each variable gadget gives a consistent truth assignment. Each clause gadget forces at least one literal in the clause to be true. Hence the formula is satisfiable.

The graph has polynomial size because each variable and each clause contributes only a constant number of vertices and edges. Therefore

$$3\text{-SAT} \leq_p 3\text{-COLOR}.$$

Since 3-SAT is NP-complete and 3-COLOR is in **NP**, 3-COLOR is NP-complete.

Write This in Exam

State membership in **NP**, then reduce from 3-SAT. Use the palette triangle T, F, B , variable gadgets forcing x_i and $\neg x_i$ to opposite truth colors, and clause gadgets that are colorable iff at least one literal is true. This gives

$$\phi \in 3\text{-SAT} \iff G_\phi \in 3\text{-COLOR}.$$

The construction is polynomial, so 3-coloring is NP-complete.